

MADERA, CA, USA

WMO: 745046

Lat: 36.988N

Lon: 120.111W

Elev: 253

StdP: 14.56

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 93242

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	30.0	32.5	22.9	17.4	49.5	26.4	20.6	46.4	21.0	54.0	18.8	52.8	2.2	90	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.5	102.0	69.7	99.6	68.9	97.3	67.8	72.8	95.7	71.1	94.2	69.5	92.3	8.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	91.6	82.4	63.0	86.9	80.9	61.1	81.2	78.7	36.5	94.8	34.9	94.1	33.6	92.4	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	16.2	13.5	DB	26.5	106.7	2.6	1.9	24.6	108.1	23.1	109.2	21.6	110.3	19.7	111.7	(4)
(5)			WB	25.1	75.3	2.5	2.2	23.3	76.9	21.8	78.2	20.3	79.4	18.5	81.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.1	47.5	51.1	55.9	60.3	67.9	75.0	80.0	78.4	74.2	64.6	54.3	47.3		
	DBStd	12.77	5.11	4.93	5.61	6.57	6.22	6.15	4.86	4.16	5.64	5.75	5.65	5.21		
	HDD50	309	111	42	12	3	0	0	0	0	0	0	23	118		
	HDD65	2383	544	389	284	170	40	4	0	0	4	77	322	549		
	CDD50	5092	32	73	196	313	554	749	930	881	726	452	152	34		
	CDD65	1692	0	0	3	30	129	303	465	416	281	64	1	0		
	CDH74	23759	0	2	126	578	2057	4361	6459	5524	3645	957	50	0		
	CDH80	13238	0	0	20	201	938	2472	3982	3304	1996	320	5	0		
(14) Wind	WSAvg	6.0	4.6	5.4	5.8	7.1	7.9	7.9	7.2	6.7	5.7	4.8	4.1	4.7		
(15) Precipitation	PrecAvg	11.2	1.8	1.8	1.9	1.2	0.2	0.0	0.0	0.0	0.2	0.6	1.7	1.5		
	PrecMax	22.1	5.7	5.3	5.1	5.5	1.8	0.4	0.1	0.3	1.6	2.0	4.3	3.3		
	PrecMin	5.6	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	4.0	1.5	1.5	1.3	1.2	0.4	0.1	0.0	0.1	0.4	0.6	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.9	73.2	82.3	91.6	99.0	103.9	106.0	104.3	101.4	93.4	79.2	66.7		
		MCWB	54.0	58.6	61.5	63.2	67.6	71.2	72.3	69.2	67.6	65.4	60.3	54.5		
	2%	DB	63.6	68.7	77.2	85.7	93.6	100.1	102.2	100.3	98.4	87.7	74.5	63.0		
		MCWB	53.1	56.8	59.7	61.8	65.6	69.3	70.3	68.6	67.1	63.4	58.5	52.8		
	5%	DB	60.4	65.2	73.3	80.9	89.8	96.8	99.7	98.3	95.2	83.1	70.6	60.0		
		MCWB	52.3	55.0	58.8	60.4	63.5	67.6	69.8	68.0	66.1	61.3	57.0	52.3		
	10%	DB	57.1	62.2	69.6	75.5	85.1	92.9	97.0	95.3	91.3	79.5	66.3	57.0		
		MCWB	51.1	53.7	57.3	58.4	61.9	66.1	68.5	67.0	65.4	60.6	55.3	51.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.1	60.6	65.1	67.4	70.3	74.7	75.6	73.2	71.9	68.0	62.5	58.6		
		MCDB	63.2	68.4	76.8	83.9	93.8	98.4	100.4	95.1	93.9	84.3	74.3	63.8		
	2%	WB	55.5	58.5	62.2	64.1	67.5	71.7	73.3	71.2	69.2	65.7	60.3	55.7		
		MCDB	60.2	66.4	73.6	81.4	89.4	95.0	96.1	94.8	92.0	81.9	71.4	59.5		
	5%	WB	53.5	56.4	59.9	61.7	65.3	69.4	71.4	69.6	67.7	63.7	58.5	53.9		
		MCDB	57.7	63.8	70.6	77.3	86.2	91.9	94.5	92.7	90.9	79.6	68.0	57.8		
	10%	WB	51.4	54.4	57.8	59.7	63.2	67.8	69.8	68.4	66.2	61.8	56.6	51.8		
		MCDB	56.6	60.8	67.8	73.9	82.7	89.7	92.9	91.1	88.6	76.6	64.6	55.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.7	20.6	23.8	26.6	30.0	32.9	33.5	33.9	32.7	28.0	23.2	18.2		
		MCDBR	23.9	25.7	30.2	34.1	36.3	36.0	34.7	36.2	36.4	33.2	29.2	22.2		
		MCWBR	15.0	15.1	14.9	13.6	13.0	11.9	11.4	11.1	11.8	13.7	15.3	13.7		
	5% WB	MCDBR	17.5	22.8	27.7	30.8	34.2	34.2	32.9	34.2	33.9	29.0	25.3	15.2		
		MCWBR	11.9	14.0	15.0	13.6	13.0	11.9	11.7	11.6	11.7	12.1	13.9	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.349	0.362	0.384	0.376	0.365	0.362	0.377	0.366	0.370	0.379	0.372	0.357		
	taud		2.370	2.344	2.278	2.333	2.388	2.397	2.374	2.382	2.380	2.318	2.289	2.334		
	Ebn at Noon		263	274	279	287	291	290	285	286	279	264	250	249		
	Edh at Noon		29	34	39	39	38	38	38	37	35	34	31	29		
(44) All-Sky Solar Radiation	RadAvg		775	1106	1570	2001	2385	2669	2583	2359	1953	1419	957	705		
	RadStd		111	127	137	156	132	97	92	71	62	82	55	97		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.69	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.90	N/A	-2.09	N/A	N/A	N/A	N/A	-245	+304	+95

Nomenclature: See separate page